

0 5 . 1

A student investigated starch hydrolysis using the enzyme amylase.

During the procedure, the student:

- treated the starch to make it soluble
- prepared 10 cm³ of different concentrations (mg dm⁻³) of starch solution
- added an identical concentration of amylase to each starch solution
- measured the time in minutes to completely hydrolyse starch.

He repeated the procedure and calculated the mean time to completely hydrolyse starch in each concentration of starch solution.

Draw a table the student could use to record all of his results.

You only need to show completed column headings.

[2 marks]**0 5 . 2**

Describe the results you would expect the student to obtain.

[1 mark]



0	5	.	3
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A competitive inhibitor decreases the rate of an enzyme-controlled reaction. Explain how.

[3 marks]

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When bread becomes stale, the structure of some of the starch is changed. This changed starch is called retrograded starch.

Scientists have suggested retrograded starch is a competitive inhibitor of amylase in the small intestine.

Assuming the scientists are correct, suggest how eating stale bread could help to reduce weight gain.

[3 marks]

